

Energy Systems Overview

Energy is the capacity to do work and is fundamental to modern civilization.

Electricity generation involves converting primary energy sources into electrical power.

Renewable energy sources include solar power, wind energy, hydroelectric power, and biomass.

Solar energy systems convert sunlight into electricity using photovoltaic cells.

Wind energy harnesses kinetic energy from wind using turbines.

Fossil fuels such as coal, oil, and natural gas are non-renewable energy sources.

Energy transmission and distribution systems deliver electricity from power plants to consumers.

Smart grids use digital technology to improve efficiency, reliability, and sustainability of power systems.

Energy efficiency focuses on reducing energy consumption while maintaining output performance.